CV1 2026 Course Syllabus

(Last updated: ~~28 Aug~~ ~~23 Sep~~ Oct 5, 2026)

General Information

- Course: Computer Vision 1 (2026/2027)
- Program: The first year of Master Artificial Intelligence
- Institution: Informatics Institute, University of Amsterdam
- Coordinator: Yen-Chia Hsu <y.c.hsu@uva.nl>
- Co-lecturer: Arnoud Visser <a.visser@uva.nl>
- Senior TAs: Fivos Tzavellos <f.i.tzavellos@uva.nl> and Arun Mukundan <a.mukundan@uva.nl>
- Refer to datanose for the time table and classroom location.
- Refer to Canvas for announcements, the link to live lectures, and other members' emails in the teaching team.

Course Description

Digital cameras have become ubiquitous in the form of consumer cameras, webcams, mobile phones, and professional cameras. These cameras yield enormous streams of data and provide the means for communication, observation, and interaction. In this course, a broad range of techniques are studied on how computers can understand the visual world of humans including image formation and filtering, features, visual information representation, learning and classification, dimension reduction, object detection and classification, object tracking, and user interaction.

Notice that the course instructor writes the syllabus from the first-person perspective. The target audience for this document is students, and thus, we use "you" to refer to the students who are reading this document.

Learning Objectives

After completing this course, the student will be able to:

- Describe the theoretical foundations of image formation, measurement, and visual analysis in computer vision.
- Apply physical-, statistical- and deep learning models to computer vision problems.
- Explain the fundamental aspects of different computer vision models and to explain their advantages and shortcomings.
- Implement and evaluate computer vision algorithms.
- Learn how to design and execute computer vision applications.

Prerequisites

This course expects you to have the following prior knowledge:

- Extensive experience with Python programming.
- General background in mathematics (e.g., linear algebra, calculus, probability theory).
- Basic knowledge of image processing and computer vision.

Course Structure

This course has 8 weeks in total. The study material is given by the lecture slides, papers, and relevant state of the art literature (all readings provided on Canvas). We also use a freely downloadable textbook (http://szeliski.org/Book/), the 2022 edition. Check Datanose for the detailed structure, time, and location.

The lectures cover the exam material, and will provide a theoretical understanding of basic Computer Vision techniques that you will apply during the lab assignments.

Seminars are given to practice the basic theory and assist you with solving exercises. These exercises are good preparation for the exam, and you should solve them as much as possible on your own without looking at the solutions.

Practical lab sessions will be held that will help you understand the practical side of computer vision tasks. The labs consist of programming tasks and open-ended questions, and is designed to provide you hands-on experience with modelling computer vision architectures.

For the seminar exercises and the labs, we will provide you the solutions, and you do not need to submit your answers. Exercises and labs are not graded, but you should still do them since they may appear in the exams.

We will use Ed Discussion on Canvas for offline conversations. This is the place to ask technical and practical questions related to the course materials, assignments, projects, or administrative matters.

Schedule Outline

Week 1

- Lecture 1: Introduction & Course Information
- Lecture 2: Camera, Image formation, and Color
- Lab 0: Introduction to Numpy & OpenCV
- Readings: Szeliski textbook section 1.1, 1.2, 2.1.1-2.1.5, 2.2, 2.3.1, 2.3.2

Week 2

- Lecture 3: Reflection, Photometric Stereo, and Image Processing
- Seminar 1: Recitations and exercises that cover prior lectures

- Lecture 4: Neighborhood Processing and Image Filtering
- Lab 1: Colour, Intrinsic Image Decomposition & Photometric Stereo
- Readings: Szeliski textbook section 3.1.1, 3.1.2, 3.1.4, 13.1.1, 3.2, 3.3, 3.4

Week 3

- Lecture 5: Local Features, Edges, Lines, and Corners
- Seminar 2: Recitations and exercises that cover prior lectures
- Lecture 6: Optical Flow, Motion and Tracking
- Lab 2: Neighborhood Processing, Low-level Filters & Applications in Image Processing
- Readings: Szeliski textbook section 7.1, 7.2, 7.4, 9.1.1, 9.1.3, 9.3

Week 4

- Lecture 7: Global Transformations and Image Stitching
- Seminar 3: Recitations and exercises that cover prior lectures
- Lecture 8: Image Classification
- Lab 3: Corner Detection, Optical Flow & Feature Tracking
- Readings: Szeliski textbook section 2.1, 8.1, 8.2, ~~13.1,~~ 6.1, 6.2.1

Week 5

- Lecture 9: Retrieval, Detection and Segmentation
- Seminar 4: Recitations and exercises that cover prior lectures
- Lecture 10: Convolution Neural Networks, Object Detection Basics
- Lab 4: Image Alignment, Image Stitching & Panorama Creation
- Readings: Szeliski textbook section 5.2.1, 5.2.2, 6.3, 6.4

Week 6

- Lecture 11: Single Shot Detection, Network Architectures
- Seminar 5: Recitations and exercises that cover prior lectures
- Lecture 12: Shape from X, Stereo, Structure from Motion
- Lab 5: Bag of Words and CNN Classification
- Readings: Szeliski textbook section 5.3, 5.4, 12.1-12.3, 13.1, 13.2
- [Update on 23 Sep: section 13.1 is moved from week 4 to here]

Week 7

- Lecture 13: Guest Lecture and Exam Q&A
    - Note that the guest lecture is not covered in the final exam.
    - This lecture will NOT be recorded.
- Seminar 6: Help students prepare exams
- ****Final exam** (check datanose for the date and location)**

Week 8

- No activity

Virtual/Physical Settings

In principle, lectures are given in person in the physical classroom. We do not livestream the lectures. If we cannot give the physical lecture due to unexpected situations, we will make an announcement on Canvas. There are two lecturers: Yen-Chia Hsu (Lecture 1, 2, 3, 4, 7, 8) and Arnoud Visser (Lecture 5, 6, 9, 10, 11, 12, 13).

Lectures will be recorded and published as soon as possible on Canvas. Only viewers with a UvAnetID can view the presentation. We will also do our best to record the lectures. However, there is no guarantee of the recording quality. For example, you may experience poor sound quality or incomplete lectures in the recorded video. If the recording equipment is unexpectedly broken or not working, we will abort the recording for the lecture. We cannot be held responsible for faulty recordings or recording equipment.

In principle, seminars and labs are in-person only and will NOT be recorded. If the course instructors or TAs cannot attend the physical classroom due to unexpected situations, they may choose to give the seminars and labs virtually. Seminars will be given by Arun Mukundan. Labs will be given by the TAs (see the Canvas home page for the TA for each study group) and supervised by Fivos Tzavellos.

Grading

Your final grade is based on one final exam (100% weight). According to the Teaching and Examination Regulations, you need to get at least 5.5 in the final grade to pass the course. There is one resit for the course, which counts as 100% weight. The most recent grade will apply in the event of a resit regardless of the outcome. This means that if you take the resit, your resit score will override the final exam grade. The resit will be at least as difficult as the final exam (which means that the resit could be more difficult than the final exam).

The base grade for each exam is 1. The base grade only applies when you attend the exam. If you miss the exam, the grade will be 0 for that exam.

After each exam, the teaching team will grade the exam and release the grades as soon as possible. After the grade is released, we will announce on Canvas how students can inspect their exam results.

Exams

Exams are onsite and will be conducted using the ANS system. You will have access to the Jupyter Notebook software for computation during the exam. You can use the question mark "?" syntax to access the IPython help to check the documentation of functions. It is not allowed to do exams at home or remotely at other locations. You will have no internet access during the exams.

Exams may contain the following types of questions to test your theoretical and practical knowledge. Check this ANS page to understand these types of questions.

- Multiple choice question (select one answer from several options)
- ~~Multiple response question (select one or multiple answers from several options)~~
  [Update on 23 Sep: we will not use multiple response questions]
- Numerical value question (provide a numerical value as the answer)

We provide mock exams for you to practice. However, due to the reason that this year's course changes the assessment method (moving from assignment with exam to only one exam), the mock exams are not representative of this year's exam, but you can still use them to practice and prepare for the final exam.

The University of Amsterdam owns the materials in this exam (e.g., the question sets), which means the materials are copyright-protected. You must keep the exam content confidential and are not allowed to copy or distribute the content in any form.

The Jupyter Notebook Coding Environment

[Update on 23 Sep: we added the coding environment clarification]

The Jupyter Notebook on the exam computers have numpy, scipy, and pandas installed.

Standard fluency. You are expected to be fluent in basic array operations, using either plain Python or numpy arrays: array creation, indexing/slicing, reshaping, elementwise operations (e.g., addition, subtraction, multiplication, division), concatenation, basic aggregation (sum, mean, variance), basic math functions (e.g., log, sine, cosine, absolute value, square root, exponential, power), and elementary linear algebra (e.g., matrix multiplication, transpose, dot product).

Syntax sheet. A reference sheet listing the call syntax for these operations is published in the syllabus in the appendix and will also be in the exam description (on your computer screen in the exam room): it is the same content in both places, and it does not count against your personal cheat sheet. You can check the signature and behavior of these functions during the exam using Jupyter Notebook's built-in question mark "?" syntax (e.g. "np.concatenate?"), which displays the docstring.

Specialized functions. Anything beyond standard fluency, such as specialized np.linalg routines (e.g., inv, det), scipy functions, or similarly non-elementary calls, will be explicitly provided in the question.

OpenCV is not installed. So, solving a question will never require writing executable OpenCV code. However, we may include OpenCV code snippets (in question descriptions or answer options) and ask you to explain what the code does. Any OpenCV function used this way will appear in the official solutions to the assignments (computer labs).

Grading for Numerical Value Questions

[Update on 23 Sep: we added the grading clarification for numerical value questions]

Numerical answers are graded as correct or incorrect. Full points will be awarded only for the correct answer. No partial point is given, regardless of how close the provided answer is to the correct answer, unless it falls within the tolerance margin defined below.

Tolerance Margin. Answers are checked against the exact, unrounded reference value (boundaries inclusive). Unless a question specifies otherwise, you need to report your final answer rounded to two decimal places, with a tolerance of ±0.01 (e.g. for the correct answer reference value 2.134324, the accepted range is [2.124324, 2.144324], including the boundary value). If a question asks for rounding to a different number of decimal places n, the tolerance is $\pm 10^{-n}$ instead (e.g., for rounding to 3 decimals, the tolerance is ±0.001). For questions with strictly integer answers (e.g., counts), no tolerance is applied: the exact integer is required as the correct answer.

Rounding. Final answers should be rounded to the precision specified in the question. Submitting additional decimal places will not be penalized, provided the value falls within the tolerance margin, as defined above.

Important. Do not round during intermediate steps. Keep exact values throughout your calculation and round only your final submitted answer, as intermediate rounding can accumulate errors and push your final answer outside the accepted tolerance margin.

Exam Grade Calculation

We use guess correction when calculating your grade on the multiple-choice ~~and multiple-response~~ questions. Details about how the ANS system uses guess correction is in this online document. We need guess correction to consider the scores that students can get by just randomly guessing. We enable the “Limit the guess correction to zero” option. The “Apply guess correction to answered questions only” option is disabled, which implies that you should always guess if you are not sure about the answer. Leaving a question blank will penalize you just as much as getting it wrong, so you should never leave a question blank.

We use the formula “1+9*obtained_points/total_points” to calculate your exam grade (defined as “mark” in the ANS system). Variable “obtained_points” means obtained points after guess correction. Variable “total_points” means the total points of the exam. For example, suppose that the exam has 50 questions, and each question is worth one point. In this case, the total points will be 50. Details about calculating the grade is in this online document.

Cheat Sheet

During the exam, you may bring **THREE A4-size cheat sheets** with you. You can choose to handwrite or print the content on both sides of the cheat sheets. After the exam, you can take the cheat sheets back with you.

The cheat sheets are designed to reduce your workload in memorizing details (e.g., math equations) and also help you prepare for exams. For example, maybe there are parts that confuse you when you study course materials, and you may put them in the cheat sheets.

Another example is that you may put bullet points that can help you recall information (e.g., some important concepts).

[Update on 23 Sep: we added this paragraph below]
We see the preparation of cheat sheets as a learning process. By writing down the concepts explicitly in the cheat sheets, you will have to prioritize information, identify the most foundational principles, and organize them in a way that makes sense to you. We believe this process can help you better understand the concepts and prepare for the exam. In this sense, students have to prepare the cheat sheets by themselves, and the course teaching team will NOT provide a unified cheat sheet to all students.

Exam Preparation

All lecture slides, seminar materials (exercises), and labs will be materials for the exam (except the guest lecture in Lecture 13). Below is a suggested exam preparation plan:

- Redo seminar exercises and labs, as well as check their answers
- Study lecture slides and the course readings
- Rewatch lecture recordings
- Prepare your cheatsheets while studying
- Do the mock exams

Note that this course has many readings (e.g., those in the “Readings” parts in the “Schedule Outline” section), and materials in the readings may also appear in the exam. The exams will be based mainly on the lecture slides, seminar materials (exercises), and labs. If a concept appears on the lecture slides, seminar materials (exercises), or labs, it may appear on the exam, and the course readings are used to support the concept with more explanations and details. However, if a concept appears in the course readings but not on the slides (also not in seminar materials and not in the labs), it will not appear on the exam.

Regarding math, as a general rule of thumb, if a math equation is on the slide, and we explain it in the lectures, seminars, or labs, it means that the equation may appear in the exam (in different ways).

Regarding math equations specifically, you need to know what they are doing at a high level. We will provide equations for complicated functions in the question description or exam instruction. But we will not provide equations for simpler things like accuracy, precision, recall, and f-score (or anything related to computing these metrics).

[Update on 5 October: we added this paragraph below]
Regarding coding, the exam can contain coding-related questions, within the boundary of the question types that we use (defined at the beginning of the “Exams” section).

- **Writing code in the exam.** If a question requires you to write code to get the solution (e.g., computing a formula or completing a specific step of a computer vision pipeline), a viable solution (among all possible alternative ways) will only require the Standard Fluency (as defined in the “Exams” section) and the code listed on the provided Syntax Sheet (in the appendix). For example, we may ask you questions that are hard to calculate by hand.

- **Reading code in the exam.** You are expected to be able to read and interpret code snippets similar to those in the exam covered materials (including lab assignment solutions, lecture slides, or seminar materials), regardless of the library/package used. For example, we may put code in the question description and ask you to select the option that best explains what the code is doing (or the expected output of the code).
- **Preparation for coding-related questions.** You can write down code snippets, formulas, or theory you have trouble recalling on your cheat sheet. To prepare for coding-related questions, we suggest re-doing the assignments (as well as checking the code that appears in the lecture slides and seminar materials) and focusing on the concepts and logic of how the code works.

[Update on 23 Sep: we added this paragraph below]
Because this is an MSc level course, exams will test your ability to synthesize knowledge, not just recall it. You should expect to see novel questions and scenarios that were not explicitly demonstrated in the lectures, seminars, or computer labs. While you will never need unrelated outside domain knowledge to solve these, you are expected to fluently apply foundational prerequisite knowledge (such as linear algebra, calculus, and probability theory) alongside the core concepts taught in this course. For example, you might be asked to solve an unseen transformation problem by synthesizing the principles of projective transformations, photometric stereo, and your assumed background in linear algebra, calculus, and probability theory.

[Update on 23 Sep: we added this paragraph below]
Some parts of the course materials (e.g., in seminar exercises and computer labs) are marked as optional. These are only for your own interest. You will not be tested on any code or advanced applications that are newly introduced in these parts. While an exam question might test a foundational concept from a lecture, seminar, or computer labs that happens to appear in an optional part, it will only test it at the level taught in the mandatory materials.

Important Notice on the Exam Day

- **VERY IMPORTANT: Anyone who arrives more than 30 minutes late cannot participate in the exam. This is the university's rule.**
- Arrive 15 minutes early to prepare for logging into the computer and ANS. Please get your login account name and password ready.
- Bring your ID cards (BOTH student ID and government-issued ID). We will check your ID cards at the exam location.
- You are only allowed to enter the exam room up to 30 minutes past the exam starting time. The purpose of the rule is to prevent fraud. This is the university's rule.
- You are not allowed to leave the exam room during the first 30 minutes of the normal exam schedule. The purpose of the rule is to prevent fraud (because students can be late to the exam up to 30 minutes).
- You can only bring pens, snacks, drinks, cheat sheets, and your ID cards to the exam room. Other things are not allowed and must be stored in the lockers, such as calculators, books, mobile phones, smart watches, coats, bags, etc.
- If you really need to go to the toilet, please raise your hand and notify the supervisors. A supervisor will need to accompany and wait for you outside the toilet.

- There are scrap papers at the exam location, and we will provide them.
- You must follow the instructions in the exam room (or outside the room) to sit in specific rows. The room may host multiple exams at the same time, and we need to locate you to check the IDs. If you sit in other rows, we may not be able to locate you, and in this case, your exam result could be invalid.
- Once you start logging in to the computer, do NOT change seats. If you change the seat and the computer logs in later using your account, the other student who comes later and sits at the desk with the computer (which is already logged in using your account) will share the same storage space for the Jupyter notebooks, which is strictly not allowed. If the login process takes too long, please raise your hand to inform the exam supervisors, and the technician in the room will help you.

Policy for Retaking This Course

For students who followed the course last year, we have changed the assessment method this year (the assignments were removed). This means that you will have to take the exam again for this year's course. We cannot carry over your scores from last year to this year.

Hygiene

Please follow the advice from RIVM (National Institute for Public Health and the Environment) regarding respiratory infections (e.g. flu, COVID-19). If you have symptoms associated with respiratory infections, please stay home.

Course Registration

For general course registration, please refer to the UvA document. The Faculty of Science handles registration procedures. The teaching team does not handle course registration matters. If you want to register for the course but you are late, or if you have problems signing up for the course, please get in touch with the following email:

- vakaanmelding-fnwi@uva.nl

Change Groups

The course study groups (the ones with letters or numbers for your seminars, such as "Group A" or "Group 1") are already assigned. After the course starts, you can still request group changes. The general rule is that you need to find someone who is willing to switch groups with you. Then, both of you need to send a message on Canvas or by email to the TAs or course coordinator to confirm your willingness to switch groups. People who follow this rule will receive higher priority in the group change.

You can also contact the TAs or course coordinator to change groups without finding a person who is willing to switch groups with you, and we will do our best to accommodate your needs. People who go this route will receive lower priority.

We cannot guarantee a successful group change in all the above cases (either following the general rule or just submitting preferences), as the teaching team needs to balance the workload among teaching assistants.

Course Attendance

This course will not track attendance (e.g., lectures, seminars, labs). We expect students to follow their own learning progress. You do not need to notify the course management team and the TAs in case of absence.

Fraud and Plagiarism

This course follows the UvA Fraud and Plagiarism Regulations. When in doubt, please consult the "Regulations Governing Fraud and Plagiarism for UvA Students" document.

Communication Principles

Class members are expected to treat others with mutual respect and appreciation regardless of any differences. It is our intent that students from diverse backgrounds and perspectives be well served by the course. In this way, we can create a safe environment for discussions.

The best ways to contact TAs and the course management team outside the lectures and seminars are via Ed Discussion on Canvas, the Inbox message on Canvas, or email. We prefer Ed Discussion as the questions can be allocated to the teaching team members based on their types. Also, other students may have the same question and thus can check Ed Discussion to get answers. For questions that are sensitive or may have privacy issues, please contact the course coordinator by email. Please keep in mind that responses from TAs and the course management team could be delayed due to weekends or holidays.

If you need quick feedback, always come to the lectures, seminars, or labs to seek support. Messages that are sent via emails, Canvas Inbox, and Ed Discussion are monitored on a weekly basis (not daily) and could be delayed.

All course announcements will be on Canvas. We expect students to monitor Canvas periodically for any changes in deadlines, or any other announcements.

For all conversations in the course (e.g., lectures, seminars, labs), please follow the "Moving the Discussion at the Speed of Trust" guidelines below.

Moving the Discussion at the Speed of Trust

- Be an active listener
- Go for honesty and depth - share what has personal meaning (with "I" statements)
- Be fully present, but step away if you need to
- Practice mutual respect
- Do not make assumptions, ask questions

- Suspend judgment
- Assume best intentions
- Invite and honor the diverse range of opinions and experiences
- Make space for all to share
- Respect the privacy of anything shared in your group
- Take care of yourself and others
- Notice your feelings of: comfort, discomfort, defensiveness, etc.
- Pause to reflect before reacting
- Expect and accept non-closure

Support for Students Wellness

If you experience mental health concerns or stressful events that can interfere with learning and daily activities (such as strained relationships, increased anxiety, substance use, feeling down, difficulty concentrating, and lack of motivation), UvA services are available. You can learn more about (confidential) mental health services available on campus in this UvA link.

As documented in the Teaching and Examination Regulations, for any personal circumstances that impact your studies (e.g., illness, disability, or family issues), contact your study advisor. Exceptions can only be made if the study advisor contacts the course management team for special arrangements, and these exceptions sometimes may require approval from the examination committee.

If you are a top-class athlete, please check this UvA page for support.

Accommodations for Students with Disabilities

If you have a disability and require accommodations, please take a look at this UvA link to request special facilities or additional resources.

Social Safety

If you experience an unsafe situation or undesirable behaviour in this course or study program, you can turn to the UvA Social Safety Support for help on how to deal with this.

Policy for Using Generative AI Tools

We allow students to use GenAI tools for seminar exercises, labs, and exam preparation. However, exams are conducted in a controlled environment without internet or GenAI access. Using GenAI comes with risks that may negatively impact your performance. GenAI tools can hallucinate information, generate wrong content, or provide fake sources/citations. Blindly relying on GenAI may lead to a misunderstanding of the course materials.

You are responsible for fact-checking and verifying all AI-generated content. The official lecture slides, course readings, and the provided answers remain the single source of truth. If

the GenAI tool generates content that contradicts the course materials, you must follow the course materials. The teaching team is neither responsible nor liable for your exam errors resulting from reliance on unverified AI-generated content.

If you are using GenAI to help you study, you can use the course materials for fact-checking by uploading them to your GenAI tool's private session that respects GDPR and the EU AI Act (e.g., using UvA AI Chat is safe). The course readings are provided for your personal use only. Please do not host, republish, or redistribute course readings to respect the license and copyright terms.

If you are not familiar with GenAI, we recommend checking the follow e-learning modules:

- AI Literacy for students
- Getting started with UvA AI Chat
- UvA AI Chat manual

If you are interested in the motivation of this GenAI policy, please check the following paper. Our opinion is largely aligned with the views in this paper. We prefer embracing its benefits so that we can jointly understand how people use it, and thus can provide us opportunities and reflections about how this kind of tool should be used or regulated in the future.

- Van Dis, E. A., Bollen, J., Zuidema, W., van Rooij, R., & Bockting, C. L. (2023). ChatGPT: five priorities for research. Nature, 614(7947), 224-226.

FAQ

Below is a list of commonly asked questions and their answers.

Q1: Can I take exams remotely?

- You have to do the exams on-site (in the physical exam room). It is not allowed to do exams remotely (e.g., at home).

Q2: What should I do if I have missed (or need to miss) an exam, such as due to medical issues or personal reasons?

- The best option is to take the resit, which is designed for these situations.

Q3: I want to change the seminar group. What should I do?

- Read the "Change Groups" section.

Q4: I need help in course registration (e.g., I register the course late). What should I do?

- Read the "Course Registration" section.

Q5: What is the cheatsheet for the exam and how should I prepare for that?

- Read the "Exams" section.

Q6: What to do if I have conflicts with my exam schedule (e.g., two exams are at the same time)?

- Contact the study adviser for solutions.

Q7: Can I reuse the partial grade from last year?

- Read the “Policy for Retaking This Course” section.

Acknowledgment

This course inherits teaching materials from Prof. Theo Gevers, Dr. Shaodi You, Dr. Martin Oswald, and Dr. Dimitris Tzionas, the previous coordinators and instructors for this course.

Appendix

Reference Coding Function Sheet for Exam

[Update on 23 Sep: we added this sheet, which will be included in the exam description]

We use “np” for numpy, “a” for numpy array objects, “pd” for pandas, and “df” for pandas DataFrame objects. You can check the signature and behavior of these functions during the exam using Jupyter Notebook’s built-in question mark “?” syntax (e.g. “np.concatenate?”), which displays the docstring.

Remember that you need to import the modules, such as

- import numpy as np
- import scipy.ndimage
- import scipy.signal
- import pandas as pd

Below are Numpy functions:

Creation	np.array, np.asarray, np.zeros / np.zeros_like, np.ones / np.ones_like, np.empty, np.eye, np.arange, np.linspace, np.meshgrid
Shape/structure	np.reshape(a, shape) / a.reshape(shape), np.transpose(a) / a.transpose() / a.T, a.flatten(), a.copy() / np.copy(a), np.stack, np.hstack, np.vstack, np.concatenate, np.tile, np.repeat, np.flip, np.diag, np.resize(a, shape) / a.resize(shape), a.astype(dtype)
Elementwise math	np.abs, np.sqrt, np.exp, np.log, np.log2, np.log10, np.sin, np.cos, np.arctan2, np.degrees, np.ceil, np.floor, np.round, np.clip(a, ...) / a.clip(...)
Aggregation	np.sum / a.sum(), np.mean / a.mean(), np.std / a.std(), np.average, np.nanmean, np.max / a.max(), np.min / a.min(), np.argmax, np.argmin, np.argsort, np.unique, np.histogram, np.gradient
Linear algebra	np.dot / a.dot(b), np.matmul, np.outer, np.linalg.inv, np.linalg.det, np.linalg.norm, np.linalg.lstsq
Logic	np.where, np.any, np.allclose, np.isnan
Random	np.random.seed, np.random.rand, np.random.randint
Constants	np.pi, np.nan, np.newaxis
Misc Numpy functions	a.item(), np.size(a) / a.size, a.view(), np.int32, np.double, np.shape / a.shape, np.deg2rad()

Below are Scipy functions:

Scipy	scipy.ndimage.gaussian_filter, scipy.ndimage.maximum_filter, scipy.ndimage.rotate, scipy.signal.convolve2d

Below are Pandas functions:

Pandas	pd.DataFrame, df.values, df.sample(n)

Below are Python built-in functions:

I/O & inspection	print(x), len(x), type(x), isinstance(x, type), hasattr(obj, name), open(path)
Iteration helpers	range(start, stop, step), enumerate(iterable), zip(*iterables), map(func, iterable), filter(func, iterable), iter(x), next(x), sorted(iterable)
Type conversion	int(x), float(x), str(x), list(x), tuple(x), set(x)
Math / aggregation	sum(iterable), max(iterable), min(iterable), abs(x), round(x, ndigits), all(iterable), any(iterable)
Misc	slice(start, stop, step), super(), format(x, spec)
For Python String s	s.split(sep), s.join(iterable), s.lower(), s.replace(old, new), s.endswith(suffix)
For Python List	list.append(x)
For Python Dictionary d	d.keys(), d.items(), d.update(other)
For Python Set s	s.update(other)